KATHERINE BRUMBERG

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Legal name: Katherine Morgan

ACADEMIC APPOINTMENTS

Assistant Professor | ***Statistics*** Jul. 2024 – Present
University of Michigan

EDUCATION

Ph.D. | ***Statistics and Data Science*** Aug. 2019 – May 2024
The Wharton School of the University of Pennsylvania

"Novel Stratification and Matching Methods in Observational Studies", *advisors: Paul Rosenbaum and Dylan Small*

Certificate | ***Teaching and Learning*** Jan. 2023
Center for Teaching and Learning at the University of Pennsylvania

B.S. and M.A. | ***Statistics and Data Science*** Aug. 2015 – May 2019
Yale University

PUBLICATIONS

7. **Brumberg, K.**, Rosenbaum, P.R. (2026+). Constructing Observational Block Designs When the Propensity Score Exhibits Limited Overlap. *Submitted.*.

6. Silber, J.H., Rosenbaum, P.R., Reiter, J.G., Ramadan, O.I., Jain, S., Hill, A.S., **Brumberg, K.**, Fleisher, L.A. (2025). Grading Hospitals Using Multivariate Matching. *Medical Care*. 63(11):817-826. https://doi.org/10.1097/mlr.0000000000002188.

5. **Brumberg, K.**, Small, D.S. & Rosenbaum, P.R. (2024). An Observational Study of the Effects of High School Football on Cognition Late in Life Using a New Matching Method. *Annals of Applied Statistics*. 18(4):3507-3527. https://doi.org/10.1214/24-AOAS1949.

   Protocol available via *Zenodo*: https://doi.org/10.5281/zenodo.8349374.

4. **Brumberg, K.**, Small, D.S. & Rosenbaum, P.R. (2024). Optimal Refinement of Strata to Balance Covariates. *Biometrics*. 80(3). https://doi.org/10.1093/biomtc/ujae061.

3. **Brumberg, K.**, Ellis, D.E., Small, D.S., Hennessy, S. & Rosenbaum, P.R. (2023). Using Natural Strata When Examining Unmeasured Biases in an Observational Study of Neurological Side Effects of Antibiotics. *J R Stat Soc Series C*. 72(2):314-329. https://doi.org/10.1093/jrsssc/qlad010.

2. **Brumberg, K.**, Small, D.S. & Rosenbaum, P.R. (2022). Using Randomized Rounding of Linear Programs to Obtain Unweighted Natural Strata that Balance Many Covariates. *J R Stat Soc Series A*. 185(4):1931-1951. https://doi.org/10.1111/rssa.12848.

1. Sinnott, J.A, **Brumberg, K.**, Wilson, K.M., Ebot, E.M., Giovannucci, E.L., Mucci, L.A. & Rider, J.R (2018). Differential Gene Expression in Prostate Tissue. *European Urology*. 74(5):545-548. https://doi.org/10.1016/j.eururo.2018.05.006.

BOOKS

2. **Brumberg, K.** (2024). Novel Stratification and Matching Methods in Observational Studies. *Proquest*. https://repository.upenn.edu/handle/20.500.14332/60102.

1. Ewens, W., **Brumberg, K.** (2023). Introductory Statistics for Data Analysis. *Springer Nature*. https://doi.org/10.1007/978-3-031-28189-1.

SOFTWARE PACKAGES

4. **Brumberg, K.**, Rosenbaum, P.R. (2026). observationalBlocks: Block Designs for Observational Studies. https://CRAN.R-project.org/package=observationalBlocks.

3. **Brumberg, K.** (2023). triplesmatch: Match Triples Consisting of Two Controls and a Treated Unit or Vice Versa. https://CRAN.R-project.org/package=triplesmatch.

2. **Brumberg, K.** (2022). optrefine: Optimally Refine Strata. https://CRAN.R-project.org/package=optrefine.

1. **Brumberg, K.** (2021). natstrat: Obtain Unweighted Natural Strata that Balance Many Covariates. https://CRAN.R-project.org/package=natstrat.

STUDENTS

Dissertation Committee Member

Jaylin Lowe — Current

Josh Wasserman — Current

PhD Research Advisor

Dili Maduabum — Current

Master's Research Advisor

Chenshuo Wu — Current

TEACHING EXPERIENCE

Professor — Fall 2024 – Present

University of Michigan Statistics Department

- "Honors Introductory Statistics" in Fall 2025, leading 2 graduate student instructors and instructing about approximately 87 students
- "Applied Regression Analysis" in Fall 2024 and Winter 2025, leading up to 5 graduate student instructors and instructing up to 200 students

Instructor of Record — Summers 2022 & 2023

Wharton Statistics and Data Science Department

- Taught the condensed six week summer course "Introductory Statistics" online

Graduate Teaching Assistant — Fall 2021 & Spring 2022

Wharton Statistics and Data Science Department

- Led recitation sections, assisted the professor with course preparation, and graded for the course "Introductory Statistics"

Undergraduate Learning Assistant — Fall 2017 – Spring 2019

Yale Statistics and Data Science Dept.

- Helped instruct students and grade for the courses "Computational Tools for Data Science," "Theory of Statistics,""Data Mining and Machine Learning," and "Data Analysis"

Honors and Awards

LSA Collegiate Fellow 2024
University of Michigan

J. Parker Bursk Research Prize 2023
Wharton School

Brown Best Student Paper Award 2022
Wharton School

Donald S. Murray Teaching Prize 2022
Wharton School

George James Doctoral Fellowship 2019
Wharton School

Graduate Research Fellowship 2019
National Science Foundation

Statistics and Data Science Senior Award 2019
Yale University

Y-Work Award for Outstanding Student Employees 2019
Yale University

Henry S. McNeil Summer Fellowship 2016
Yale University

Yale Club of Boston Scholarship 2015, 2016, 2018
Yale University

Lori Laitman Rosemblum Scholarship 2015
Yale University

External Conferences and Presentations

Triples Matching; the Effects of HS Football on Cognition Late in Life Oral presentation
Joint Statistical Meetings Aug. 2025

Triples Matching; the Effects of HS Football on Cognition Late in Life Poster and flash talk
IMS New Researchers Conference July-Aug. 2025

Triples Matching; the Effects of HS Football on Cognition Late in Life Poster presentation
American Causal Inference Conference May 2025

Optimal Stratification to Address Selection Bias in Observational Studies Oral presentation
University of Notre Dame Jan. 2024
Villanova University Dec. 2023
Wake Forest University Nov. 2023
University of Michigan Nov. 2023
Hamilton College Nov. 2023

Swarthmore College Nov. 2023

Vassar College Nov. 2023

Reed College Nov. 2023

Optimal Refinement of Strata to Balance Covariates Oral presentation

Joint Statistical Meetings Aug. 2023

American Causal Inference Conference May 2023

Lawrence D. Brown PhD Workshop Nov. 2022

Obtaining Unweighted Natural Strata that Balance Many Covariates Using Randomized Rounding of Linear Programs Speed presentation

Joint Statistical Meetings Aug. 2021

Academic and Community Service

Judge Mar. 2026
Michigan Student Symposium for Interdisciplinary Statistical Sciences

Committee Member Mar. 2026
MAS Admissions Committee, University of Michigan Statistics Department

Founder and Co-organizer Feb. 2026 – Present
Interdisciplinary Causal Inference Working Group, University of Michigan

Faculty Mentor Sep. 2025 – Present
First Year PhD and Bridge Students, University of Michigan Statistics Department

Committee Member Sep. 2025 – Present
Seminar Committee, University of Michigan Statistics Department

Committee Member Sep. 2025 – Present
Graduate Curriculum Committee, University of Michigan Statistics Department

Mentor Jun. – Jul. 2025
Big Data Summer Institute at the University of Michigan

Committee Member Jan. 2025
PhD Admissions Committee, University of Michigan Statistics Department

Committee Member Sep. 2024 – April 2025
Undergraduate Curriculum Committee, University of Michigan Statistics Department

Peer Reviewer Aug. 2023 – Present
Journal of Educational and Behavioral Statistics, Journal of Causal Inference, Springer, Epidemiology, Journal of Quantitative Analysis of Sports, Journal of Computational and Graphical Statistics

President Sep. 2019 – Aug. 2023
Wharton Society for the Advancement of Women in Business Academia (WSAWBA)

- Led the organization which aims to support women in the Wharton doctoral programs

Pen-pal Sep. 2019 – Jun. 2023
Letters to a Pre-scientist

- Wrote letters to a middle school student at a underresourced school to broaden awareness of what STEM professionals look like and serve as a source of inspiration

Mentor Sep. 2022 – Dec. 2022

Wharton Pre-Doctoral Directed Reading Program

- Developed a reading plan with an undergraduate student and met weekly to discuss the reading

Advisory Board Member Sep. 2021 – May 2022

Wharton Pre-Doctoral Directed Reading Program

- Matched undergraduate students with graduate members for the program and helped to develop the new program

Committee Member Jan. 2022 – Apr. 2022

Wharton Statistics and Data Science Quinquennial Review

- Gathered doctoral student feedback for the quinquennial departmental review

Mentor Jan. 2021 – Jun. 2021

Graduate School Mentoring Initiative

- Mentored a first generation low income undergraduate student interested in applying to graduate school

Co-President Sep. 2015 – May 2019

Splash at Yale

- Led the nonprofit educational outreach organization which brings 1000 middle and high school students to campus to take hour long seminars taught by Yale students

Tutor

New Haven Reads Feb. 2016 – May 2017

- Volunteered as a tutor for underresourced students reading below grade level in the New Haven school district